REVIEW REPORT · SLITTER 2

Cutting Slitter 2 changeover time

The best approach, from 6 viewpoints and 5 meetings

Based on: "Slitter 2 Work Efficiency Analysis" (12 Oct 2026) · utilisation 23%, 373 min stopped, same-spec changes 184 min, spec changeovers 143 min

Before you read — This is not real meeting minutes. It is a **simulated review** run by Claude, based on the 12 Oct PLC/video analysis and the nature of each role (the 3 Slitter 2 crew, Plant Manager, CEO, Sales Director). Each statement describes "the concerns and views likely to come from that seat". Please confirm and adjust them with the people themselves in a real meeting.

The 5 meetings

Meeting 1
Facts
Where does the time go? Agreed: "it's the order of work, not the people"

Meeting 2
Same-spec changes
Move banding off the line. Two-step banding trial

Meeting 3
Spec changeovers
Knives first when the line stops + spacer kits prepared in advance

Meeting 4
Orders, crane, lunch
Orders fixed by 15:00 the day before, same-spec grouping, crane rules

Meeting 5
Final agreement
Fixed roles, 2-week trial, targets and how to measure

The best approach (conclusion)

"Prepare while the line runs what we now prepare after it stops" — with each person owning a fixed part.

1. **Fixed changeover roles + a sequence card** — Melven = lead, run, thread · Young-chul = next-job prep & knife change · Nad = unloading & banding · Mark = crane & labels
2. **Two-step banding** — only a temporary band at the recoiler; final banding off the line while the next coil runs
3. **Spacer kits** — next job's spacers set up in order while running; knives changed the moment the line stops
4. **Orders fixed by 15:00 the day before** — same specs grouped; tomorrow's first job set up before leaving
5. **Stop-reason logging + weekly PLC report** — results visible in numbers

Cost: **small** (stands, kit cart) · **no extra staff** · Target: utilisation **23% → 30%** in one month, 35% once established

1 Six viewpoints — what each person worries about and wants

How each seat is likely to read the analysis report. The meetings were run to close the gaps between them.

Melven · Operator

Slitter 2 equipment handling · line lead

Concern Everything piles up at once during a changeover, and who does what has to be decided verbally every time. He cannot leave the control panel for long.

Wants The next job's materials already waiting, so the knife change can start the moment the line stops.

Suggests A picture-based changeover sequence card (Korean & English). If each task is on the card, fewer instructions are needed.

Young-chul · Spacers

Slitter 2 spacer work

Concern A 20-strip job takes real time to set spacers. The spacer bench is off-camera, so set-up time can look like "waiting".

Wants Getting the next job early, a place to pre-set spacers, and spacer set-up recognised as his own responsibility.

Suggests A **kit cart** holding the next job's spacers in order. Prepared while running, they can go straight in after the stop.

Nad · Helper

Slitter 2 helper (unloading & banding)

Concern If banding moves off the line, the work lands on him. Not sure he can finish 20 final bands alone before the next unload.

Wants Stands and banding tools in one place; someone to help when busy (Mark).

Suggests First time how long one band takes; if it is too tight, Mark helps with labels and final banding.

Plant Manager

Plant operations · Slitter 1 equipment handling

Concern If both lines share the crane, Slitter 1 may wait while Slitter 2 pre-hangs coils during its run. An off-line banding spot must not block aisles (safety).

Wants Crane priority rules, a marked banding area, and a rule for splitting work between Slitter 1 and 2.

Suggests Move some 7–10-strip jobs to Slitter 1. Bring Slitter 1's "prepare in advance" habits to Slitter 2 where they fit.

CEO

Management · investment decisions

Concern Can it be done without more staff or big equipment? Can the effect be seen in numbers?

Wants No-cost steps right away; small spending once results show. Weekly utilisation figures.

Suggests Decide on numbers after a 2-week trial. Big items like a spare arbor only after the trial.

Sales Director

Sales · customers

Concern Separate banding by destination and thickness/weight markings are customer requirements and cannot be dropped. Changing the order must not delay deliveries.

Wants More capacity means more orders can be accepted. Customer packing standards written down.

Suggests Fix the next day's jobs by 15:00 the day before and group same customer / same spec. Write a packing-standard sheet per customer.

Where the viewpoints clash

- **Banding off the line** → the line gets faster but Nad's workload grows (Meeting 2)
- **Hanging the next coil while running** → good for Slitter 2, may clash with Slitter 1 on the shared crane (Meeting 4)
- **Preparing spacers early** → only possible if job information arrives early → sales / order timing (Meetings 3–4)
- **Investment** → the floor wants equipment, the CEO wants a no-cost trial first (all meetings)

Meeting 1 Facts — where does the time go?

Agenda: share the 12 Oct report, agree on the numbers, decide where to start · Attendees: all six

CEO 23% utilisation means the machine stands for more than 6 hours a day. Was it just this day, or is it always like this?

Analysis The 6-week average is about 31%. 12 Oct was a bad day, but changeovers taking longer than running is the normal pattern.

Melven · Operator Right — the 9 afternoon changes had no knife change and still took 20 minutes each. Most of it is unloading and banding.

Young-chul · Spacers The report shows nobody from 11:37 to 11:50, but the spacer bench is not on camera. That whole time should not be counted as waiting.

Plant Manager This is not about who was idle. The problem is that preparation only starts after the line stops.

Sales Director Separate banding by destination on the Yuchang job is a customer requirement. We cannot remove the banding itself.

Nad · Helper Right now all the banding is finished in front of the recoiler before unloading. The line stays stopped the whole time.

Open points

- The real reason for the 14 minutes without movement — off-camera spacer work or waiting — is **not confirmed**
- If the report reads as "who was late", the floor will get defensive

Decisions — Meeting 1

- The cause is treated as **the order of work, not the people**. The data is not used for individual evaluation.
- **Log stop reasons for 2 weeks** (unload / band / knives & spacers / coil loading / waiting / material) — floor-screen buttons or a paper sheet
- Get the 12 Oct afternoon footage and a view of the spacer bench to time each step.
- Start with the biggest loss: **same-spec changes (banding)** → Meeting 2

Meeting 2 Same-spec change 20 → 12 min — how to take banding off the line

Agenda: cut the 20-minute stop per coil when the same spec runs back to back, as in the afternoon

Option	What	Effect	Cost	Verdict
A. As is	Finish final banding at the recoiler, then unload	None	None	Baseline
B. Two-step banding	At the recoiler, only a **temporary band** to hold it → unload onto stands → **final banding & labels** while the next coil runs	**~8 min per coil**	2 stands etc., small	**Adopted (trial)**
C. Add staff	One more person dedicated to banding	Yes	Labour cost	CEO against · on hold
D. Simpler packing	Agree a simpler banding method with the customer	Yes	None	Slow · in parallel

Nad · Helper B is good, but a coil only runs 6–7 minutes and then the next unload comes. I am not sure I can finish 20 final bands alone in that time.

Analysis The next coil's run (6.5 min) plus the following change (target 12 min) gives about 18 minutes. At 40 seconds per band, that is about 13 minutes. The real time needs measuring.

Plant Manager Stands go next to the recoiler exit, outside the forklift aisle. Mark the area on the floor, and check every time that the temporary band holds.

Melven · Operator One temporary band is enough. If unloading gets faster, I go straight into threading the next coil.

CEO No to C. B with a couple of stands — try it now. If the numbers show it works, I will support more.

Sales Director I will talk to Yuchang about D, but it takes time, so B first is right. I will hand over a packing-standard sheet per customer for the label items.

Young-chul · Spacers On same-spec runs I have no knife work. I can use that time to prepare the next spec's spacers, or help with final banding when it is busy.

Decisions — Meeting 2

- **Two-step banding trial** — one day, the next time the same spec runs back to back
- Final banding: Nad. **Mark supports with labels and final banding**; Young-chul can help on same-spec runs
- **Time one band** before the trial → adjust support if there is not enough slack
- 2 stands and floor marking (Plant Manager); packing-standard sheet per customer (Sales Director)

Meeting 3 Spec changeover 48 → 25 min — knives and spacers

Agenda: bring spec changeovers back to the usual level (22 min)

Option	What	Cost	Verdict
A. Knives first at the stop	When the line stops, unloading (Nad) and knife change (Young-chul & Melven) start **at the same time**	None	**Adopted · now**
B. Spacer kits	Set up the next job's spacers in order on a kit cart while running	Kit cart, small	**Adopted**
C. Spare arbor	A second arbor swapped in as a whole	Large	Check machine first · on hold
D. Day-before set-up	Before leaving, set knives, spacers and first coil for tomorrow's first job	None	**Adopted**

Young-chul · Spacers If I know the next spec early, I can line up the spacers in order while the line runs. The knives sit on the arbor, so the actual change has to wait for the stop — but with a ready kit it is much faster.

Melven · Operator When the line stops I will start unloading and knife change together. Today knife work only starts after the previous product is cleared, so it drags.

Plant Manager First check we have enough spare spacers — setting up the next set separately needs that many. C is too expensive for now.

CEO Do A, B and D now. Buy the kit cart. We will revisit C after the trial results.

Sales Director For B to work, job information must come down early. I will fix the next day's jobs by 15:00 the day before.

Nad · Helper While they change knives, I take unloading and clearing the recoiler. Nobody waits for anybody.

Still to check

- Spare spacers — enough to set up the next set separately? (if not, add a few of the most-used widths)
- Spare arbor — machine design and cost

Decisions — Meeting 3

- **"When the line stops, knives first"** — unloading and knife change in parallel
- **Spacer kits** are Young-chul's responsibility. Kit cart purchase and space (Plant Manager)
- **Day-before set-up** — about 20 min after the last job, Young-chul & Melven
- Check spare spacers (Plant Manager & Young-chul)

Meeting 4 Orders, crane, lunch — what blocks the line from outside

Agenda: the operating rules that must change for Meetings 2–3 to work in practice

Sales Director I will group same customer / same spec in the orders. Urgent deliveries are the exception, and I will tell production why when that happens.

Plant Manager Slitter 1 already runs 12–14 jobs a day, so there is little slack. Only some 7–10-strip jobs can move. Moving too many just adds changeovers on Slitter 1.

Plant Manager The crane needs rules: ① the line in a changeover has priority ② pre-hanging during a run only when the other line is not in a changeover ③ if they clash, Mark decides the order.

CEO What about staggered lunch breaks so the line keeps running?

Melven · Operator With one of three away, unloading and banding fall behind. I also worry about safety. Not now.

Nad · Helper If a changeover lands just before lunch, we finish the knives, take lunch, and run straight away after.

Young-chul · Spacers Day-before set-up is impossible if the last job ends late. Please plan the last job's end time when issuing orders.

Topic	Decision	Owner
Order timing	Next day's jobs **fixed by 15:00 the day before** → to production (same specs grouped)	Sales Director
Line split	Move 7–10-strip jobs to Slitter 1 only when it has slack (adjusted weekly)	Plant Manager
Crane use	Line in changeover first · pre-hanging only when the other line is not changing over · Mark resolves clashes	Plant Manager & Mark
Changeover before lunch	Staggered lunch **on hold**. If a spec change starts after 11:30, finish knives & spacers before lunch → run at 13:00	Melven
Day-before set-up	Plan orders so the last job ends before 16:50 (time for set-up)	Sales Director & Plant Manager

Remaining disagreements

- **Staggered lunch** — the CEO wants it; the floor objects on staffing and safety. Revisit with trial results (utilisation)
- **Number of cranes and routes** — this review assumes both lines share a crane. Needs checking

Meeting 5 Final agreement — who, when, and how we measure

Agenda: turn Meetings 1–4 into roles, schedule and target numbers

Roles during a changeover (printed on the sequence card)

Who	While the line runs	When the line stops
Melven Line lead	Running · check next job order · decide when to call the crane	Call unloading and knife change to **start together** → join knife change → thread & run
Young-chul Set-up prep owner	Prepare the **next job's spacer kit** · check knives · (same-spec runs) help final banding	**Change knives & spacers at once** · before leaving, set up tomorrow's first job
Nad Unloading & banding	**Final banding** and tidying of the previous coil (off-line stands)	Temporary band → unload onto stands → clear the recoiler
Mark Shared	**Pre-hang** the next coil (crane rules) · print labels · help final banding	Crane for coil loading

Support roles

Who	What
Plant Manager	Space and floor marking for stands and kit cart, crane rules, spare-spacer check, 15-min floor check once a week
Sales Director	Jobs fixed by 15:00 the day before, same specs grouped, packing-standard sheet per customer, packing talks with Yuchang
CEO	Approve small purchases (stands, kit cart), review at 2 weeks and 1 month, decide big items (spare arbor, staggered lunch)

Targets (measured automatically from the PLC every week)

Measure	12 Oct	Usual (6 wk)	In 2 weeks	In 1 month
Utilisation	23%	31%	28%	**30%+**
Same-spec change (each)	20 min	13.5 min	15 min	**12 min**
Spec changeover (each)	48 min	22 min	30 min	**25 min**
First coil running by	08:55	-	08:45	**08:40**

Targets go "back to usual first, then beyond". Daily numbers swing with the job mix (spec changes, strips, coil length), so they are judged as weekly averages.

Result The best approach and the plan

Across the five meetings, every "work faster" idea dropped out. What remained is **moving preparation earlier and giving each person a fixed part**. All six viewpoints could accept it for three reasons:

Floor crew (3)
Who does what is fixed, so fewer verbal instructions; prep work is recognised as each person's own job

Plant Manager
Crane and safety-area rules come first, and Slitter 1 is not overloaded

CEO · Sales
Starts with no new staff or big spending, and customer requirements (separate banding, markings) are kept

Schedule

When	Action	Owner
This week 13–16 Oct	Hand out role & sequence cards (Korean/English) · start "knives first when the line stops" · start stop-reason log · time one band · check spare spacers	Melven · Plant Manager · Nad
Next week 19–23 Oct	Set up stands and kit cart · start fixing jobs by 15:00 the day before · **two-step banding trial** on a same-spec day · start day-before set-up	Plant Manager · Sales Director · all
After 2 weeks week of 26 Oct	First review with the weekly PLC report · update the sequence card with floor feedback	All (30-min meeting)
After 1 month 2nd week of Nov	Review against targets · revisit spare arbor, staggered lunch, simpler packing	CEO leads

Risks and responses

Risk	Response
Banding piles up on Nad	Time banding before the trial · Mark and Young-chul support · rebalance after 2 weeks
Temporary band slips → quality/safety	Temporary band standard (count, position) on the card · check final banding is done
Crane clashes (Slitter 1 waits)	Changeover-first rule · Mark resolves clashes · check crane waiting after 2 weeks
Feeling of "being watched"	Make clear in the first meeting the numbers are for fixing the sequence, not rating people · share results with the floor first
Orders not fixed the day before	Flag missed days in the weekly report · weekly sales–production check

Check before the real meeting — ① the real reason for the 14 minutes (11:37–11:50) ② number of cranes and routes for the two lines ③ spare spacers ④ time for one final band ⑤ the scope of Yuchang's packing and marking requirements. The details above may change depending on these five points.